MIL-H-83267A(USAF)
20 September 1971
SUPERSEDING
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29 April 1970

MILITARY SPECIFICATION

* HOISTING UNIT, AIRCRAFT MAINTENANCE HLU-213/E;
HELICOPTER SERVICING, 8,000 POUND CAPACITY

1. SCOPE

* 1.1 This specification covers a 8,000-pound capacity, helicopter servicing, aircraft maintenance hoisting unit, designated HLU-213/E.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	Strapping, Steel, Flat and Seals
RR-W-410	Wire Rope and Strand
TT-P-664	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
TT-V-121	Varnish, Spar, Water-Resisting
UU-T-81	Tags, Shipping and Stock
VV-L-800	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-P-40	Packaging and Packing of Hand Tools
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-S-207	Sulfuric Acid, Electrolyte, Packaging, Packing, and Marking for Shipment and Storage of

* FSC 1730

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MIL-[illegible]	Plates, Identification, Instruction, and Marking, Blank
MIL-[illegible]	Test, Air [illegible] Equipment, [illegible]
[illegible]	[illegible]
MIL-C-[illegible]	Casting, Classification and Inspection of
MIL-H-[illegible]	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-M-[illegible]	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-[illegible]	Air Transportability Test Procedures, General Specification for
MIL-[illegible]	Hose Assembly, Rubber, Hydraulic, High Pressure 15,000 PSI
MIL-H-[illegible]	Hose Assembly, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-E-[illegible]	Engine, Preparation for Shipment and Storage of
MIL-G-10924	Grease, Automotive and Artillery
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-E-16298	Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of
MIL-C-17504	Coating Compound, Acrylic, Clear
MIL-H-19253	Hooks, Hoist
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base

STANDARDS

Military

MIL-STD-12	Abbreviations for use on Drawings, Specifications, Standards, and in Technical Documents
MIL-STD-17	Mechanical Symbols
MIL-STD-100	Engineering Drawing Practices
MIL-STD-[illegible]	Marking for Shipment and Storage
MIL-STD-[illegible]	Identification Marking of US Military Property
MIL-STD-[illegible]	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-[illegible]	Mobile Electric Power, Engine Generator Set Family, Characteristic Data Sheets
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals

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MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods
MS27609	Electrical Connectors for Test Equipment and General Utility
MS28774	Retainer, Packing Backup, Single Turn, Tetrafluorethylene
MS51336	Lunette-Coupler, Drawbar, Ring

DRAWING

Sikorsky Aircraft

65010-00050 Interface Control - General Arrangement, HH-53 Helicopter

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Board of Fire Underwriters

Pamphlet 70 National Electrical Code (Electric Wiring and Apparatus)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The hoisting unit, hereafter designated hoist, shall consist of at least the following major components:

	Item or System Name	See Requirement
	Frame structure	3.7.1
	Lift system	3.7.2
	Running gear	3.7.3
	Electrical system	3.7.4
*	Hydraulic system	3.7.5.

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4. Protective treatment. Materials used in the construction of the hoist [illegible] against [illegible] compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.2 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are to be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.3 Standard steels. Steels complying with ASTM or AISI standards shall be utilized in the design and fabrication of the hoist and shall be corrosion resistant or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered during storage or normal service.

3.4.3.1 Heat treatment. Heat treatment of steel materials shall be in accordance with MIL-H-6875.

3.4.4 Castings. Castings shall be designed and classified in accordance with MIL-C-6021.

3.4.5 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

* 3.5 Design and construction. The hoisting unit shall be designed as a self-contained, air-transportable unit capable of removing and installing the dynamic components from the HH-53B/C aircraft during maintenance operations at an established base or remote site. The hoist shall be designed to support a rated load of 8,000 pounds and shall be capable of accomplishing the maintenance functions designated herein independent of the airframe. The design shall eliminate all unnecessary weight and shall reflect an effort toward weight-strength optimization. Overall design load factors shall include accelerations encountered during air transport in accordance with MIL-A-8421; ground accelerations, vibrations, and other applicable conditions in accordance with MIL-M-8090; and ground handling or any other factors, as applicable, which govern the maximum loads to be encountered by the hoist during normal service.

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3.5.1 The hoist shall be designed and constructed to assure that all parts and components are so assembled as to be incapable of working loose during service. The hoist shall be designed to withstand the structural loads, strains, vibrations, and other conditions incident to operation, shipping, and storage.

3.5.2 Functional design. The hoist shall be designed for positioning at appropriate points along either side of the aircraft, as shown on Drawing 65010-00050, to individually service the main and tail propulsion components and shall have access to all main and tail components without interfering or contacting any part of the aircraft.

3.5.3 Interface capability. The hoist shall provide an interface capability with at least the following aerospace ground equipment items and aircraft:

a. Sikorsky aircraft Drawing 65010-00050

b. Lifting hook to component slings (see 3.7.2.3.1)

c. Hoist to floor tiedown (see 3.5.8.3)

d. Winch attach points for ramp loading (see 3.5.9)

e. Standard type AF tow vehicle.

3.5.4 Reliability. The hoist shall have a minimum mean-time-between-failures (MTBF) of not less than 120 cycles with a reliability of not less than 0.990 for a one-cycle mission at 0.9 confidence.

3.5.5 Maintainability. The hoist shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data

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determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general purpose tools and equipment normally available commercially. Use of special purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimize the number of tools required for maintenance by design practices that use standardized tools and fasteners to the practicable extent

f. [illegible] shall be designed so that it can be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.6 Human engineering. At least the following recognized principles of human engineering shall be given consideration in the design of the hoist:

a. Human space limitations for operation and maintenance

b. The intellectual, physical, and psychomotor capabilities of the intended user

c. Visual and auditory perceptual requirements

d. Arrangements and readability of control and instrument displays

e. Safety factors covering minimization of potential human error in the operation and surveillance of the equipment

f. Comfort factors relative to efficiency of operation

g. Effects of climatic and mechanical service conditions on the ability of personnel to perform as required.

3.5.7 Welding. Welding shall be accomplished by operators certified in accordance with MIL-T-5021, class A.

3.5.8 Air transportability. The hoist shall be designed and constructed to meet the air transportability requirements specified in MIL-A-8421, while in the folded configuration. The hoist shall meet those requirements pertaining to floor loading, ramp negotiation, acceleration loads, tiedown provisions, and markings.

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* 3.5.8.1 Weight and envelope constraints. The maximum gross weight of the hoist shall not exceed 10,000 pounds. During air transportation, the hoist shall not exceed the floor strength limits illustrated on figure 1. The envelope dimensions of the hoist shall not exceed the cargo compartment dimensions of the HH-53B/C aircraft as illustrated on figures 2 and 3.

3.5.8.2 Ramp loading. The hoist shall negotiate a ramp angle of 13.5° and fit within the dimensional confines of the cargo bay entrance of the HH-53B/C aircraft as illustrated on figures 2 and 3.

3.5.8.3 Tiedown provisions. The hoist shall be equipped with tiedown fittings in accordance with MIL-STD-209. The position and combined structural restraint shall be in accordance with MIL-A-8421. The location and cone of action of tiedown fittings in the HH-53B/C aircraft are shown on figures 4 and 5 respectively.

3.5.8.4 Lifting rings. Lifting rings shall be provided for lifting the hoist in the folded locked configuration for external air transport using the HH-53B/C aircraft cargo hook. The lifting rings shall support 300 percent of the gross weight of the hoist and shall be so positioned as to permit equal distribution of the load to each lifting ring.

3.5.9 Winch loading. Provisions shall be made for winch loading and unloading of the hoist in the folded locked configuration into and from the cargo compartment of the HH-53B/C aircraft.

3.5.10 Cables or wire rope. The use of cable or wire rope on the hoist shall be kept to a minimum, and when used shall conform to RR-W-410.

3.5.11 Operating and service controls. All operating and service controls shall be so arranged that they are readily operable by personnel wearing heavy arctic gloves and clothing.

3.6 Performance

3.6.1 Environmental conditions. The hoist shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Operating temperatures ranging from -65° to +120°F

b. Storage temperatures ranging from -80° to +160°F

c. Precipitation (rain or snow) and ice formation

d. Relative humidity up to 100 percent

e. Exposure to salt fog

f. Fungus growth as encountered in tropical areas

g. Dust as encountered in desert areas.

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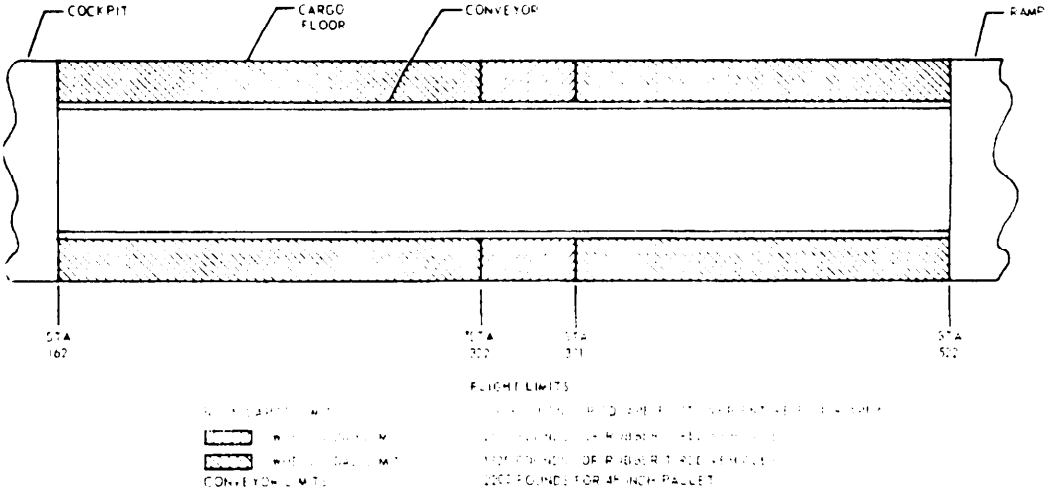


FIGURE 1. Floor Strength Limits (Flight Limits)

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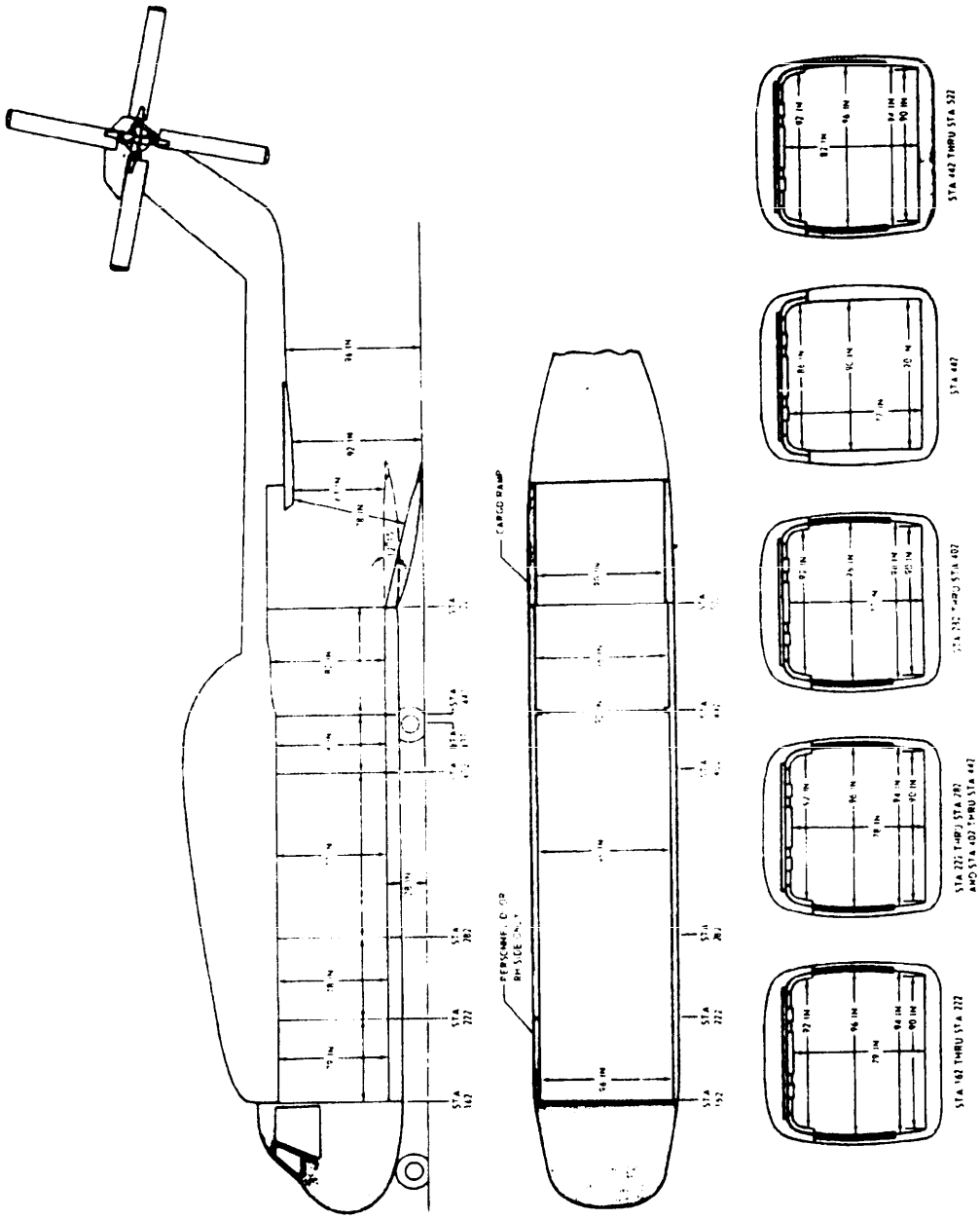


FIGURE 2. Cargo Compartment Dimensions

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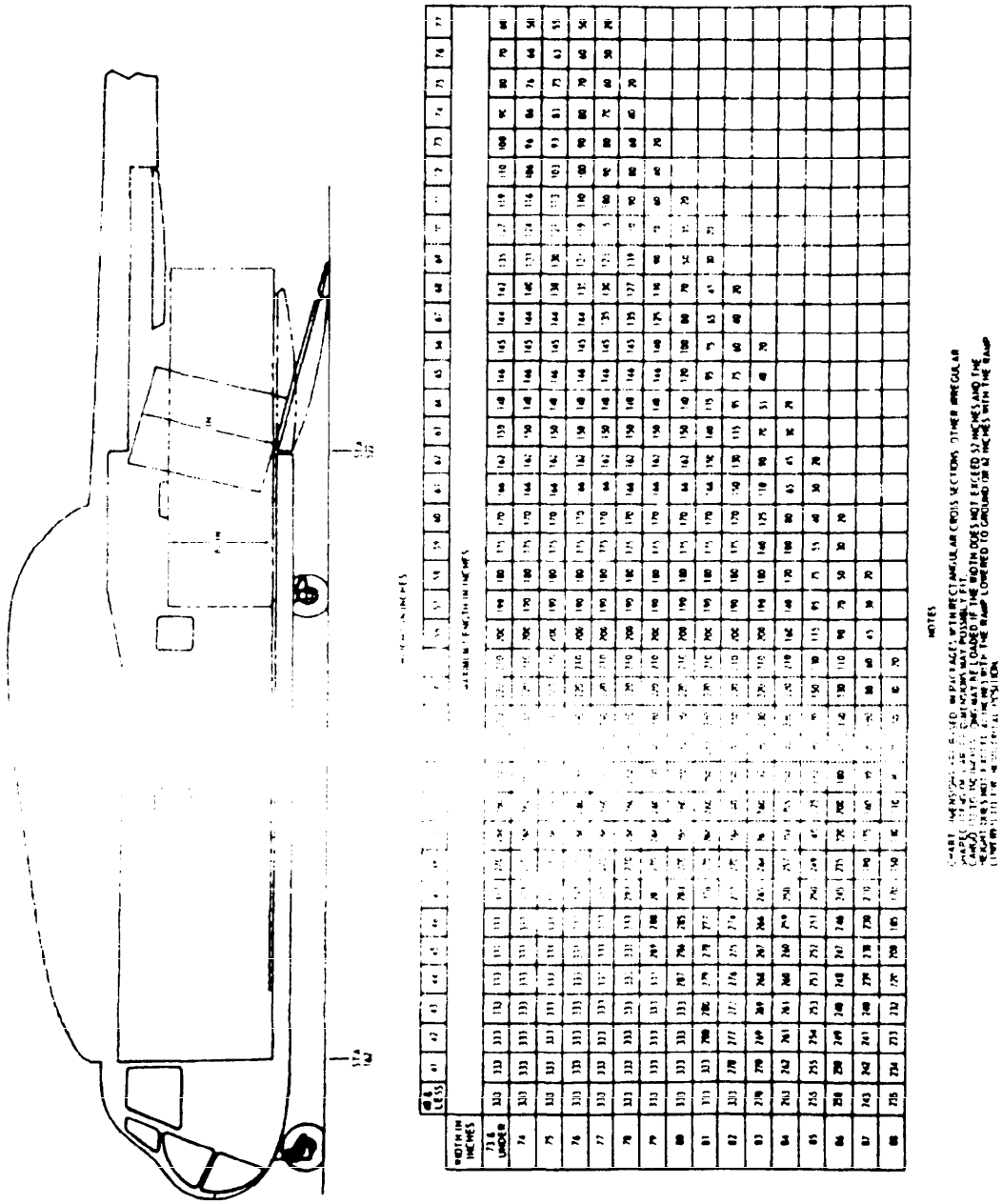


FIGURE 3. Cargo Size [illegible] for Ramp Loading-Ramp Lowered to Ground

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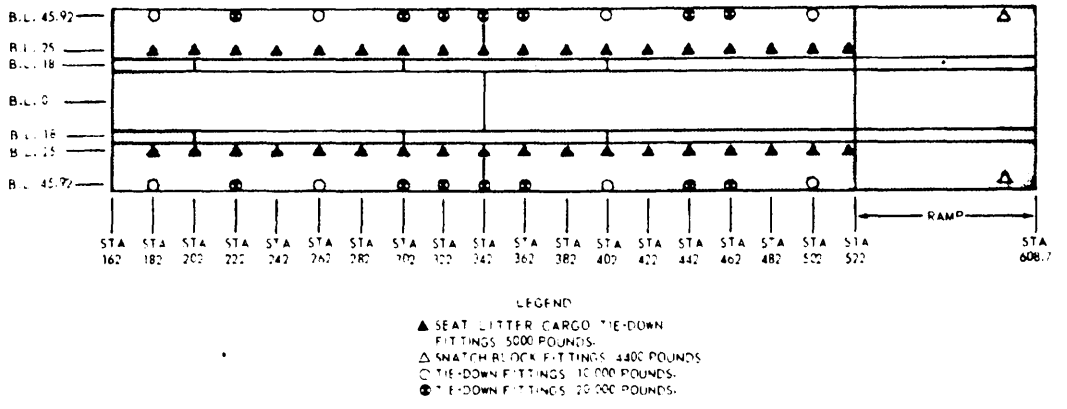


FIGURE 4. Location of Tiedown Fittings

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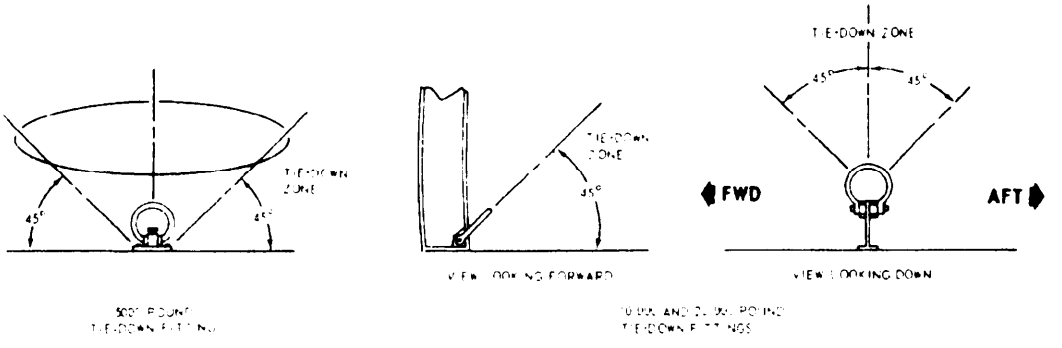


FIGURE 5. Tiedown Fittings - Cone of Action

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* 3.6.2 Mobility. The hoist shall withstand the shock and vibration of mobile operation over flight line and uneven off-runway surfaces within the confines of the towing speed limits as specified herein.

* 3.6.3 Proof load. The hoist shall withstand a static proof load of 16,000 pounds at the lifting hook (see 4.6.5.1).

3.7 Details of components

3.7.1 Frame structure. The frame structure shall be capable of folding into a configuration suitable for storage, towing, and air transport. The frame design shall incorporate telescoping features or hinged joints provided with a suitable power system to actuate the erection and retraction of the frame. The frame shall be so designed that it can be erected and retracted by one person. Locks shall be provided to secure the frame components in the retracted position during storage and air transport. The locks shall be manually operated without the use of handtools.

3.7.1.1 Stabilizing features and footpads. The frame design shall include outriggers, extension legs, and other stabilizing features, as necessary, to provide a stable base for the hoist while operating under the conditions specified herein. Footpads shall be provided at appropriate locations to aid in support of the bearing load to the terrain surface. The footpad activation mechanism shall be capable of hand operation without the use of tools and capable of being locked in place at any one setting.

3.7.1.2 Erection and retraction controls. Controls for the operation of the erection and retraction mechanism shall be located to insure the safety of operating personnel against movement of the frame components during erection and retraction operations. The controls shall be arranged and clearly identified to avoid confusion and shall be located to prevent inadvertent actuation or damage during normal handling operations.

3.7.1.3 Safety locks. Positive mechanical safety locks shall be provided on the frame to secure the frame when in the erected operational configuration. If hydraulic components are used on the frame erection and retraction system, safety locks shall be designed to relieve the load from the hydraulic components during operation of the hoist. The safety locks shall be manually operated without the use of handtools.

3.7.2 Lift system. The lift system shall be capable of removing and installing the dynamic components specified herein from and to its place on the aircraft, to and from a trailer or workstand positioned alongside the aircraft. The lift system shall be designed to impart a lateral, transverse, and vertical freedom of movement to the lifting hook.

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3.7.2.1 Lateral swing. The lift system shall be capable of rotating in a plane parallel to the slope upon which it rests with sufficient lateral swing to permit the lifting and lowering of a component to a trailer or workstand alongside the aircraft. The lateral swing mechanism shall be powered by a power system capable of swinging a component of rated load through a sloping arc with the hoist resting on a maximum positive or negative slope of 5°. The lateral swing mechanism shall be capable of stopping at any point along its arc travel and include a fine adjustment capability using its powered source. Positive mechanical stops shall be provided at the end points of the swing travel.

3.7.2.2 Transverse carriage. The transverse carriage shall have sufficient fore and aft travel to individually service the main and tail propulsion components with the hoist positioned on one side of the aircraft. The transverse carriage shall be powered by a suitable power system and shall be capable of stopping at any point along its travel. The transverse carriage shall be capable of fine adjustment in its horizontal travel using its powered source. Positive mechanical stops shall be provided at the end points of the carriage travel.

3.7.2.3 Vertical lift. The vertical lift mechanism shall be capable of lifting from the ground line to a minimum height of 24 feet at the central radius of the lifting hook. The lift mechanism shall operate a lifting hook through the transverse carriage assembly powered by a suitable power system and capable of stopping at any point along its travel. The lift mechanism shall be capable of fine adjustment in its vertical travel using its powered source. If a hydraulic system is used, a positive mechanical safety lock shall be designed into the lift system to relieve the load on the hydraulic components and shall be designed to be engaged and disengaged at any position along the range of the lift. The safety lock shall be manually operated without the use of handtools.

3.7.2.3.1 Lifting hook. A swivel type lifting hook equipped with a sling [illegible] shall be provided at the interface between the vertical lift mechanism and the component slings. The hook shall be in accordance with MIL-H-19253 and shall be capable of supporting a rated load of 8,000 pounds with a safety factor of three.

3.7.2.4 Lift system controls. Controls for the operation of the lift system shall be designed as a hand-held electrically controlled unit. The control cable shall be of sufficient length to permit the use of the control unit by maintenance personnel located on the ground or topside of the aircraft. The control unit shall operate independently the movement and fine adjustment of the lift system in the vertical, transverse, and lateral swing directions. The controls governing the movement of the lift system in each of the three

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14. WORK TELEPHONE NUMBER (Include

directions shall be so arranged and clearly identified to avoid confusion to operating personnel. A mounting bracket shall be located on the frame structure to provide for storage and protection of the control unit and lead cable while not in use.

3.7.3 Running gear. The running gear shall meet type II, group C requirements of MIL-M-8090 with the exceptions and additions specified herein.

3.7.3.1 Self-propelled wheel drive. A self-propelled wheel drive shall be incorporated on the hoist running gear for the purposes of aiding in the final positioning of the hoist alongside the aircraft, and limited mobility in the immediate servicing area about the aircraft. Power for the wheel drive shall be supplied by either a wheel coupled hydraulic or electric motor. The motor shall include a control system capable of accomplishing fine positioning of the hoist alongside the aircraft.

3.7.3.2 Parking brakes. A mechanically actuated brake system shall be provided on the running gear. The parking brake shall be capable of restraining the hoist on a 13.5° ramp incline (see 3.5.8.2). Controls for operating the parking brakes shall be conveniently located on the hoist frame and shall have provisions for positive locking of the parking brakes in the engaged and disengaged positions.

3.7.3.3 Steering and towing. A towbar-controlled steering mechanism shall be provided on the hoist for normal towing and ground mobility requirements as specified herein. The towbar shall have a lunette eye conforming to MS51336 and shall be designed to hinge or be easily detachable for storage when not in use.

3.7.3.4 Towing speed. The maximum towing speed of the hoist in the erected configuration shall be 5 mph. In the retracted/folded configuration, the maximum towing speed shall be 10 mph.

* 3.7.4 Electrical system. The electrical system used on the hoist shall be designed for use in hazardous locations in accordance with the provisions of class I, division 2, group D of Pamphlet 70. All components of the electrical system shall be approved by Underwriters' Laboratories, Incorporated for use in class I, division 2, group D hazardous locations (see 6.3).

* 3.7.4.1 Electric power supply unit. The electric power supply shall be a self-contained portable unit capable of satisfying all the electric power requirements of the hoist and selected from those listed in MIL-STD-633. Selection of an electric power unit shall be with the approval of the procuring activity (see 6.2). The power unit shall be mobile and movable by two persons.

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The hoist frame shall include a provision for mounting and securing the power unit to the frame during storage, towing, or air transport. The power unit shall be capable of operation while secured to or positioned away from the hoist frame.

* 3.7.4.1.1 Protective cover. A weatherproof cover shall be provided to enclose the portable power unit to protect the engine and generator during storage against adverse weather and environment.

* 3.7.4.2 Cable and connectors. A 50-foot, heavy duty, oil-proof, electric cable shall be provided on the hoist for connecting to the power unit or permanent facility electric power receptacles. The power input connector plug shall be in accordance with MS27609 for the applicable voltage and amperage. The applicable mating receptacle shall be provided on the power supply unit. A storage facility shall be provided on the hoist to store the cable when not in use.

3.7.4.3 Electrical equipment. All electrical wiring, switches, solenoids, connectors, etc, used on the hoist shall incorporate explosion-proof (vapor-proof) characteristics in accordance with class 1, division 1, group D of Pamphlet 70.

3.7.5 Hydraulic system. If hydraulic components are used on the hoist, they shall include but need not be limited to: hydraulic cylinders, valves, reservoir, lines and fittings, filter, and controls. Hydraulic cylinder packing seals shall be of the O-ring type conforming to MIL-P-5516 with V-type teflon backup rings conforming to MS28774. The hydraulic system shall be designed for use with hydraulic fluid conforming to MIL-H-5606, and shall be furnished with a reservoir filled with the proper amount of fluid. The system shall be designed and installed in accordance with the best commercial practice.

3.7.5.1 Hydraulic lines and fittings. Nonflexible metal tubing shall be used in the hydraulic system where flexing is not encountered during operation. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, best visual inspection for failures, and maximum protection from damage during operation. Flexible hose shall be used where flexing of hydraulic lines is encountered and where the installation of components requires a flexible coupling. Hydraulic hose assemblies shall be in accordance with MIL-H-8790 or MIL-H-8795, depending upon hydraulic system pressure requirements. Hose assembly installations shall be consistent with the minimum bend radii specified for the size hose used.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.9 Electromagnetic interference. The hoist and all installed components, including the power unit, shall comply with the interference requirements specified in MIL-STD-826, method 7001.

3.10 Finishes and protective coatings. Cleaning, painting, and finishing shall be in accordance with MIL-STD-808. The exterior color shall be yellow, film designation DG.

3.11 Markings

* 3.11.1 Nameplates, instruction plates, transportation and general data plates, and placards. All nameplates, instruction plates, transportation and general data plates, and placards shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required. Abbreviations shall be in accordance with MIL-STD-12. Mechanical symbols shall be in accordance with MIL-STD-17. All such plates shall be attached by metal screws or rivets.

* 3.11.1.1 Tiedown data plates. Placards depicting tiedown instructions for air transport shall be provided.

* 3.11.1.2 Lifting rings. Lifting rings shall be clearly identified as such by stenciled markings adjacent to the rings.

* 3.11.2 Operational markings and stenciling. Operational markings and stenciling shall be in accordance with MIL-STD-808.

* 3.11.2.1 Towing speed. The towbar shall be marked as follows:

MAXIMUM TOWING SPEED: FOLDED 10 MPH
ERECTED 5 MPH.

* 3.11.2.2 Load rating. The following note shall be stenciled in bold letters in a conspicuous location:

RATED LOAD: 8,000 Pounds
DO NOT OVERLOAD.

* 3.11.3 Operating instructions. Where applicable, brief operating and precautionary instructions shall be permanently affixed on or near the control panel or other points as necessary. The instructions shall be clear, concise, and adequate to enable operation without damage to the system or injury to operating personnel. Precautionary instructions shall be so designed and located that the operator cannot fail to notice them before controls, valves, and switches are operated.

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129. The nomenclature shall be:

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification accordance with MIL-STD-130.

3.13 Workmanship. The hoist, including all parts and accessories, shall be constructed and finished in a manner which reflects the highest quality workmanship. Particular attention shall be given to neatness and thoroughness of marking of parts an assemblies, and freedom of parts from burrs and sharp edges.

3.13.1 Surface cleaning. Prior to the application of surface treatments, cleaning shall be accomplished as specified herein with materials and processes that have no deleterious effects on the metal surface and by means of a procedure that is satisfactory for the application of the surface treatment required.

3.13.1.1 Preparatory cleaning. The hoist shall be thoroughly cleaned of loose, spattered, or excessive weld scale and flux; rust and metal chips; oil, greases, oxides; and other foreign matter after assembly. Burrs and sharp edges, as well as resin flash that may crumble, crack, etc, shall be removed.

3.13.1.2 Multiple-pass welds. When a weld is builtup by two or more beads or passes, preceding beads shall be cleaned free of scale and flux by chipping or scraping and brushing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the hoist shall be classified as follows:

a. Preproduction tests . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

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4.3 Test conditions

4.3.1 Standard test conditions. Unless otherwise specified herein, all tests shall be conducted under room conditions with the temperature at 25° ±10°C, the relative humidity at 90 percent or less, and the barometric pressure at 28 to 32 inches Hg.

* 4.3.2 Mockup component. A dead weight load of 8,000 pounds equivalent to the configurational dimensions of the combined main rotor head and gear box shall be provided and utilized for test purposes. The mockup component shall be handled by attachment fittings simulating those of the actual component slings.

4.3.3 Data. During the testing specified herein, sufficient data shall be recorded at proper intervals to insure compliance of the hoist with the performance requirements specified. Recorded data shall include the date, time, and ambient temperature at which the test is conducted as well as the data needed to accurately calculate the performance rates, efficiencies, etc, to be determined by the specific tests.

4.3.4 Failure during test. Characteristics exhibited by the hoist during any of the tests specified herein that would cause the hoist to fail any of the requirements while in service shall constitute a failure and shall be the basis for rejection.

4.4 Preproduction testing (see 6.2)

.1 Test sample. One hoist shall be subjected to the tests specified in 4.3.

* 4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity. Any required certifications shall be furnished with the test report (see 6.3).

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

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c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests

b. Sampling test.

4.5.1 Individual tests. Each hoist shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Operational tests.

4.5.2 Sampling test. One hoist shall be selected at random from every 10 or fraction thereof produced and subjected to 10 percent of the test specified in 4.6.8.

4.5.2.1 Rejection and retest. When one hoist selected from a production run fails to meet the specification, hoists still on hand or produced later shall not be accepted until the extent and cause of failure has been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

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4.5.2.1.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the hoists on hand or produced later shall not be made until it is determined that all hoists meet all the requirements of the specification.

4.5.3 Defects in hoists already accepted. The investigation of a test failure could indicate that defects may exist in hoists already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The hoist shall be examined to determine conformance to the requirements of this specification with respect to dimensions, quality of workmanship, materials, design and finishes, markings, and visible defects, and the functional operation of all components, assemblies, accessories, and controls. Particular attention shall be given to leakage in any hydraulic system used, and the routing and protection of hydraulic lines.

4.6.2 Operational tests. The hoist shall be operated under the following conditions to insure that the various components and capabilities comply with the requirements specified herein. During this performance demonstration, all components shall be operated according to the instructions provided with the hoist.

.2.1 Component installation and removal. Using the portable power unit secured in the stored position on the hoist frame, the hoist frame shall be erected from the folded to the operational configuration and secured. All stabilizing devices and footpads shall be set in place. The portable power unit shall then be detached from the hoist frame, positioned 50 feet from the hoist, and connected to the hoist. The hoist shall lift the mockup component to its maximum vertical travel, move through the entire range of the transverse carriage, swivel through the entire range of the lateral swing mechanism and lower the mockup component to the ground. The portable power unit shall be replaced and locked on the hoist frame. The hoist frame then shall be retracted and locked into the folded configuration. This test shall be conducted on 5° negative sloping terrain or simulated as such with the forward end of the hoist (outward end of transverse carriage travel) at a lower altitude than the aft end.

4.6.3 Mobility. The hoist shall be tested for type II, group C mobility in accordance with MIL-M-8090, with the following exceptions:

4.6.3.1 Folded configuration. In the folded configuration, the hoist shall be towed 20 miles over flight line apron or ramp type terrain at a speed of 10 $\pm$1 mph. The hoist also shall be towed 20 miles over graded gravel terrain at a speed of 10 $\pm$1 mph.

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4.6.3.2 Erected configuration. In the erected operational configuration, the hoist shall be towed 20 miles over flight line apron or ramp type terrain at a speed of 5 ±1 mph.

4.6.3.3 Parking brake. The performance of the parking brake shall be demonstrated on the 13.5° incline.

4.6.3.4 Hoist positioning. The self-propelled wheel drive shall be demonstrated by fine positioning of the erected hoist alongside the aircraft. This test shall be conducted on both paved hardtop and level grassy terrain.

4.6.4 Air transportability. The hoist shall be tested for air transportability in accordance with MIL-A-8421.

4.6.5 Static proof load tests

* 4.6.5.1 Frame and components. A static proof down load of 16,000 pounds shall be applied to the lifting hook with the frame in the erected locked configuration, the vertical lift mechanism at its maximum height above the ground, the transverse carriage at its outermost point, and the lateral swing mechanism at the midpoint of its travel. If any hydraulic system is used, the safety locks shall be engaged to relieve the load from the hydraulic components. The proof load shall be maintained for 3 minutes without evidence of leakage, instability, or permanent deformation of any component or part during or after completion of the test.

4.6.5.2 Lifting rings. It shall be demonstrated that the hoisting rings and associated structure will support a proof load of 300 percent of the gross weight of the hoist.

4.6.6 Environmental tests. The hoist or critical components thereof shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810:

a. High temperature - Method 501, procedure I

b. Low temperature - Method 502, procedure I

c. Salt fog - Method 509, procedure I

d. Dust - Method 510, procedure I

e. Humidity - Method 507, procedure I

f. Fungus - Method 508, procedure I.

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* 4.6.7 Electromagnetic interference. The hoist and all installed components, including the electric power supply unit, shall be tested for electromagnetic interference through all operations in accordance with MIL-STD-826, method 7001.

4.6.8 Endurance tests

4.6.8.1 Operational tests. With the frame in the erected, locked configuration, the hoist shall demonstrate for 50 cycles the capability to lift the mockup component from the ground to its maximum vertical travel, move through the entire range of the transverse carriage, swivel through the entire range of the lateral swing mechanism, and lower the mockup component to the ground. Twenty-five cycles shall be performed using the receptacle of the available local power while the remaining 25 cycles shall be performed using the portable power unit.

4.6.8.2 Erection and retraction. The frame erection and retraction mechanism shall be operated through 50 complete cycles with the safety locks engaged in the erected and folded configurations. Twenty-five cycles shall be performed using the receptacle of the available local power while the remaining 25 cycles shall be performed using the portable power unit.

4.6.8.3 Leakage inspection. If any hydraulic components are used on the hoist, they shall be inspected for leakage during this test. Fluid lines, fittings, pumps, cylinders, and other hydraulic components shall show no leakage. However, if pumps, cylinders, or other equipment have packed glands that can be tightened during the endurance tests to stop leakage that may have occurred, such adjustments will be acceptable.

4.6.9 Serviceability. The hoist shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given to maintenance with a minimum number of tools, maintenance with general-purpose tools and equipment, servicing and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of decontamination residue, dirt, snow, ice, etc, that may hinder servicing and operation.

4.6.10 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

4.6.11 Maintainability evaluation. The hoist shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, etc, that may hinder servicing and operation.

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4.6.11.1 Preventive maintenance, servicing, and unit replacement. Maintenance tests shall be conducted to determine the adequacy and ease of performing preventative maintenance, servicing, and unit replacement. The frequency of servicing required for the various components shall be determined by study of existing supplier's recommendations and by servicing required during the preproduction tests. Interferences or obstructions to servicing shall be noted. Any t .dency of the hoist to pick up mud, snow, and ice which interferes with servicing and operation shall be noted. Removal and installation of major components and assemblies to demonstrate ease of replacement shall be accomplished. Particular attention shall be given to time required, difficulties encountered, and the tools and skills required in conducting maintenance, servicing, and replacement.

* 4.7 Inspection of preparation for delivery

* 4.7.1 Materials and processes. All materials and processes involved in the preparation for delivery shall be inspected to determine conformance to requirements of section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be level A or C, as specified (see 6.2)

* 5.1.1 Level A Preservation and packaging for type I mobile, type II crated shipments

* 5.1.1.1 Disassembly and matchmarking

* 5.1.1.1.1 Disassembly. Disassembly of the hoist shall be the minimum necessary to safeguard parts vulnerable to damage. Protruding parts that increase cubage which can be removed and reassembled without special tools shall be removed. The bolts, nuts, screws, pins, and washers removed shall be placed with mating parts and secured to prevent loss.

* 5.1.1.1.2 Matchmarking. Removed parts and mating parts shall be matchmarked to facilitate reassembly. Matchmarking identification shall be on cloth shipping tags conforming to UU-T-81, type A or on metal tags and the tags attached to mating parts. The cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be so attached to the parts as to prevent damage to the preservation on the parts.

* 5.1.1.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in table I. Preservatives, when applied, shall not come in contact with rubber impregnated parts which may be adjacent to metal surfaces receiving the application of preservatives.

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TABLE I. Preservatives

Type	Specification	Description
P-1	MIL-C-16173, grade 1	Thin film preservative (hard drying, cold application)
P-2	MIL-C-16173, grade 2	Thin film preservative (soft film, cold application)
P-9	VV-L-800	Very light preservative oil (cold application)
P-11	MIL-G-10924	Grease, automobile and artillery

* 5.1.1.1.4 Cleaning and drying. Exterior surfaces of the hoist shall be free of dirt, dust, or other contaminating materials. Surfaces to which application of a preservative is specified shall be cleaned and dried in accordance with any applicable process and procedure of MIL-P-116.

* 5.1.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces of components of the hoist requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. Unprotected metal information plates shall be coated with varnish conforming to TT-V-121, MIL-C-17504, or equal.

* 5.1.1.1.6 Self-propelled wheel drive. If self-propelled wheel drive is not a completely sealed unit, the unit shall be removed and packaged method 1A16 or IIA in accordance with MIL-P-116. The method of preservation will be determined by the type of unit supplied. Method 1A16 will be used for the hydraulic unit and method IIA for the electric unit. The unit container shall be PPP-B-601, PPP-B-621, or MIL-B-26195 overseas type, style optional. Strapping shall be in accordance with the appendix to the applicable box specification.

* 5.1.1.1.7 Power supply

* 5.1.1.1.7.1 Engines. Engines, component parts, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, method I, type II or III, as specified (see 6.2). Batteries, cables, and electrolyte shall be prepared as specified therein and as follows: Electrolyte shall be packaged in accordance with MIL-S-207.

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* 5.1.1.1.7.2 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298, alternate method as specified for assembled machines.

* 5.1.1.1.8 Openings. Openings created by removing the units in 5.1.1.1.6 and 5.1.1.1.7 shall be sealed with cover and gasket compatible with component part.

* 5.1.1.1.9 Winch and derrick assemblies. Winch gearcase and other gear driven units shall contain gear lubricant conforming to requirements of applicable lubrication, filled to operating level. Hydraulic systems shall contain operational hydraulic fluid filled to operating level. When operating surfaces of hydraulic piston are exposed, the piston shall be coated with grease conforming to MIL-G-10924 and overwrapped with barrier material conforming to type II, grade A, class 2 of MIL-B-121 rewound with tape conforming to type III, class 1 of PPP-T-60. When hydraulic piston is retracted, exposed surface of piston shall be coated with type P-1 preservative. Hydraulic controls shall be secured in neutral position.

* 5.1.1.1.10 Braking system. Exterior unpainted or threaded surfaces such as cables, clevises, and linkage of parking brakes shall be coated with type P-1 preservative.

* 5.1.1.1.10.1 Interior surface of brake drum. Interior surface of brake drum, including brake cams, anchor pins, adjusting screw and braking surface, shall be coated with a thin film of primer conforming to TT-P-664. Care shall be exercised to prevent primer from contacting brake linings and rubber impregnated parts. A warning tag, bearing the following information shall be securely attached in a conspicious location:

BRAKE DRUMS PRESERVED: DO NOT APPLY BRAKES WHEN VEHICLE IS BEING MOVED. MOVE VEHICLE WITH RIGID TOW BAR. ADJUST BRAKES BEFORE PLACING IN OPERATION.

* 5.1.1.1.11 Maintenance tools. Maintenance tools shall be prepared and packaged in accordance with PPP-P-40.

* 5.1.1.1.12 Technical publications. Technical publications for each hoist shall be packaged in accordance with MIL-P-116, method 1C.

* 5.1.1.1.13 Wheels, axles, and bearings. When wheels and hubs are removed from axles for type II shipments in crate, the interior bearing surface of the hubs and exterior bearing surface of the axles shall be coated with type P-2 preservative. The coated surface of the axles shall be wrapped with barrier material conforming to type II, grade A, class 2 of MIL-B-121 and secured in place with tape. Openings of the hubs shall be covered with

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barrier material conforming to type II, grade A, class 2 of MIL-B-121 and secured with tape. Bearings removed from the wheels shall be preserved and packaged in accordance with method 1A of MIL-P-116, applying P-11 preservative and a preliminary wrap of barrier material conforming to type II, grade A, class 2 of MIL-B-121.

* 5.1.1.1.14 Chassis lubrication. All grease fitting and parts lubricated by the fittings shall be charged to capacity with a lubricant containing a rust inhibitor and have a capacity of retention in the fitting or in the parts when subjected to prolonged storage and high temperature conditions. The lubricant shall be in accordance with the manufacturers lubrication instructions. All operating mating surfaces of components, such as levers, latches, control linkage, locking pins, shafts, and pedal linkage, shall be coated with P-2 or P-9 preservative. Grease lubricated parts, not provided with grease fittings, shall be preserved with P-11 preservative.

* 5.1.1.1.15 Parts removed by disassembly. Unit packages (excluding power supply and self-propelled wheel drive units) and unpackaged parts removed by disassembly, accessories, and repair parts shall be intermediate packaged in nail wood boxes conforming to class 2, style optional of PPP-B-621 or in wood cleated-plywood boxes conforming to overseas type, style optional of PPP-B-601. These containers shall be so secured to the chassis as to prevent damage to the chassis.

* 5.1.2 Level C Preservation and packaging for type I mobile and type II crated shipment. The preservation and packaging requirements specified for this level are intended to provide adequate protection for immediate use, domestic shipments from supply source to the first receiving activity. This level may conform to the suppliers commercial practice when such practice meets the requirements of this level.

* 5.2 Packing

* 5.2.1 Packing shall be type I mobile or type II crated, as specified (see 6.2).

* 5.2.1.1 Levels A, B, and C

* 5.2.1.1.1 Type I mobile. The hoist, preserved and packaged as specified, shall be shipped without further packing.

* 5.2.1.2 Type II crated. The hoist, preserved and packaged as specified, shall be packed in a sheathed wood crate constructed, closed, and strapped in accordance with type I, class 1 or 2, style A or B of MIL-C-104.

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* 5.2.1.2.1 Arrangement, blocking, bracing, and anchoring. Tires and wheels, when removed from the hoist, shall be secured to the base of the crate or to the unit by bolting or strapping. Strapping shall conform to QQ-S-781, size as applicable. Cushioning dunnage shall be placed between tires and wheels and supporting surface and under all strapping to prevent damage to the tires and wheels. Blocking, bracing, and anchoring shall be in accordance with the appendix of MIL-C-104 and in accordance with MIL-STD-1186. Pneumatic tired wheels shall be blocked clear of floor of crate and shall not be load bearing.

* 5.3 Marking. Packaged shipping container and unboxes (mobile) hoists shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The HLU-213/E hoist is intended for use as a maintenance tool for the removal and installation of the dynamic components from the HH-53B/C aircraft.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. That MIL-M-008090E (USAF), dated 18 April 1966, or later issue be used in conjunction with this specification

c. Provisions for selection and approval of the electric power unit to be used (see 3.7.4.1)

d. Location and conditions for preproduction testing (see 4.4)

e. Certifications to be submitted with preproduction test report (see 4.4.2 and 6.3)

f. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2)

g. Selection of packing type I mobile type II crated (see 5.2)

h. Whether engines are to be preserved type II or III (see 5.1.1.1.7.1).

* 6.3 Certifications

* 6.3.1 Electrical certification. The contractor should submit proof that the electrical system and all of its components, which will be supplied under this

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specification meet the requirements of Pamphlet 70 and the standards of Underwriters' Laboratories, Incorporated for use in class I, division 2, group D hazardous locations. The label or listing of the Underwriters' Laboratories, Incorporated will be accepted as evidence of conforming to the specification requirements.

6.3.1.1 In lieu of the label or listing, the contractor may submit independent proof satisfactory to the procuring activity that the equipment conforms to the requirements of Pamphlet 70 and the standards of Underwriters' Laboratories, Incorporated with regard to fire and casualty hazards. Compliance with these requirements does not absolve the contractor from complete compliance with the other requirements of this specification.

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 84

Preparing activity:
Air Force - 11

Project No. 1730-F153

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD, FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 66 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

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